习题9.1 1,2,5(1)(3)(4),7,10,11,12
习题9.2 3,4,5,7,8. 习题9.3 偶数题目
习题9.4 题目1,2,5,9，每道大题任选两道小题， 本节其余题目基本上都非常经典，建议全做（至少做 n-3 道）；
习题9.5 2,4,6,8,9; 习题9.6 4,6,7. 习题9.7 1-5大题每道题任选一道小题,10 题

9.1

9.1.1

(1)

$$\int_a^b \left| x - \frac{a+b}{2} \right| \mathrm{d}x = \frac{(a-b)^2}{4}. \tag{1}$$

(2)

$$\int_a^b \sqrt{(x-a)(b-x)}\,\mathrm{d}x = \int_a^b \sqrt{-\left(x - \frac{a+b}{2}\right)^2 + \frac{(a-b)^2}{4}}\,\mathrm{d}x = \frac{\pi(a-b)^2}{8}. \tag{2}$$

9.1.2

Suppose $f \in \mathcal{R}([a,b])$ prove that

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^n f\left(a + \frac{k}{n}(b-a)\right). \tag{3}$$

Proof: Let $x_k^{(n)} = a + \frac{k}{n}(b-a)$ then for all n, $x_0^{(n)}, \cdots, x_n^{(n)}$ is a partition of $[a,b]$ and $\frac{b-a}{n}\sum_{k=1}^n f(x_k^{(n)})$ is a Riemann sum. Since $x_{k+1}^{(n)} - x_k^{(n)} = \frac{b-a}{n} \to 0$, the sum converges to the integral $\int_a^b f(x)\,\mathrm{d}x$.

9.1.5

Determine the integrability of f on $[0,1]$:
(1) $f(x) = -\mathbf{1}_{\mathbb{Q}} + \mathbf{1}_{\mathbb{Q}^C}$: non-integrable.
(3) $f(x) = \operatorname{sgn}\left(\sin\frac{\pi}{x}\right)$, $f(0) = 0$: integrable.
(4) $f(x) = \left\{\frac{1}{x}\right\}$, $f(0) = 0$: integrable.

9.1.7

Suppose $f \in \mathcal{R}([a,b])$, and f, g agree on all but finitely many points. Prove that $g \in \mathcal{R}([a,b])$ and $\int_a^b f\,\mathrm{d}x = \int_a^b g\,\mathrm{d}x$.
Proof: g is continuous a.e. since it is continuous at all continuities of f except for finitely many points. $f - g = 0$ except for $x_1, \cdots, x_n$ so $\int_a^b f - g\,\mathrm{d}x = \sum_{k=0}^n \int_{x_k}^{x_{k+1}} f - g\,\mathrm{d}x = 0$.

9.1.10-11

Suppose $f \in \mathcal{R}([a,b])$ prove that $e^f \in \mathcal{R}([a,b])$. If $f \geqslant c > 0$, $\log f \in \mathcal{R}([a,b])$.
Proof: f is bounded implies e^f is bounded on $[a,b]$. If f is continuous at x then e^f is also continuous at x, so e^f is continuous a.e. Hence $e^f \in \mathcal{R}([a,b])$.
$\log f$ is analogous.

9.1.12

If f is bounded on $[a,b]$, and its set of discontinuities D_f has only finitely many limit points, then $f \in \mathcal{R}([a,b])$.
Proof: Suppose the limit points of D_f are $x_1, \cdots, x_n$, then for any $y \notin \{x_1, \cdots, x_n\}$ there is an $\varepsilon(y) > 0$ such that $B(y, \varepsilon(y)) \cap D_f \subset \{y\}$. For any $\delta > 0$, $\bigcup_{y \neq x_i} B(y, \varepsilon(y)) \cup \bigcup_{i=1}^n B(x_i, \delta)$ is an open covering of $[a,b]$. Since $[a,b]$ is compact, there is a finite sub-covering $\bigcup_{j=1}^m B(y_j, \varepsilon_j) \cup \bigcup_{i=1}^n B(x_i, \delta)$ of $[a,b]$. Hence for any $\delta = 1/n > 0$, only finitely many points of D_f lie outside $\bigcup_{i=1}^n B(x_i, \delta)$, denote them by A_n. Note that $D_f \subset \bigcup_{n \geqslant 1} A_n \cup \{x_1, \cdots, x_n\}$, so D_f is countable, and $f \in \mathcal{R}([a,b])$.

9.2

9.2.3

If $f^2 \in \mathcal{R}([a,b])$, then $|f| \in \mathcal{R}([a,b])$.
Proof: $|f|$ is continuous at all continuities of f^2.

9.2.4

Suppose $f \in \mathcal{R}([a,b])$, if $f > 0$ then $\int_a^b f \, \mathrm{d}x > 0$.
Proof: For any partition of f, the lower Darboux sum is strictly positive, so the integral is positive.

9.2.5

Suppose $A \subset \mathbb{R}$. If A is a closed null set, prove that χ_A is integrable on any closed interval and the integral is zero.
Proof: $\chi_A = 0_{\mathbb{R}}, a.e.$ so the integral is zero.

9.2.7

Suppose $f, g \in \mathcal{R}([a,b])$. If f, g disagree on only a null set, prove that their integral are the same.
Proof: We show that if $f = 0, a.e.$ then $\int_a^b f \, \mathrm{d}x = 0$:
Suppose $|f| \leqslant M$ on $[a,b]$, then for all $\varepsilon > 0$, $\{x : f(x) \neq 0\}$ can be covered by intervals $\bigcup_{i=1}^n (l_i, r_i)$ with length less than $\delta = \varepsilon / M$. Then $\int_a^b f \, \mathrm{d}x = \sum_{i=1}^n \int_{l_i}^{r_i} f \, \mathrm{d}x \leqslant M\delta \leqslant \varepsilon$ and likewise $\int_a^b f \, \mathrm{d}x \geqslant -\varepsilon$. Therefore $\int_a^b f \, \mathrm{d}x = 0$.

9.2.8

Suppose $f \in \mathcal{R}([a,b])$, g is bounded and disagree with f only on a null set, show that $g \in \mathcal{R}([a,b])$.
Proof: The union of two null sets are still null sets, and if f is continuous at x and $f(x) = g(x)$, then g is continuous at x.

9.3

9.3.2

Give an example of F differentiable on $[a,b]$ but F' not integrable on $[a,b]$.
Solution: Consider $F : [0,1] \to \mathbb{R}, x \mapsto x^2 \sin\left(1/x^2\right), F(0) = 0$, then $F'(0) = 0$ but $F'(x) = 2x \sin\left(1/x^2\right) - \frac{2}{x} \cos\left(1/x^2\right)$ is unbounded near $x = 0$, hence F' is not Riemann integrable.

9.3.4

Calculate
(1)

$$\int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}. \tag{4}$$

(2)

$$\int_a^b \sin x \, \mathrm{d}x = \cos a - \cos b. \tag{5}$$

(3)

$$\int_0^\pi \cos^2 x \, \mathrm{d}x = \frac{\pi}{2}. \tag{6}$$

9.3.6

Suppose $f \in C([a,b])$ and $\int_a^b f^2 \, \mathrm{d}x = 0$, prove that $f \equiv 0$.
Proof: Otherwise if $f(x_0) \neq 0$, suppose $x_0 \neq b$ and let $\varepsilon = |f(x_0)|/2$, then there exists $\delta > 0$ such that $|f(y)| > \varepsilon \forall y \in [x_0, x_0 + \delta]$. Hence $\int_a^b f^2 \, \mathrm{d}x \geqslant \int_{x_0}^{x_0+\delta} f^2 \, \mathrm{d}x > \varepsilon^2 \delta > 0$ a contradiction. Hence $f \equiv 0$.

9.3.8

Suppose $f \in C^1([a,b])$ and $f(a) = 0$, prove that

$$\int_a^b f^2 \, \mathrm{d}x \leqslant \frac{(b-a)^2}{2} \int_a^b |f'|^2 \, \mathrm{d}x. \tag{7}$$

Proof: Note that

$$f(x)^2 = \left(\int_a^x f' \, \mathrm{d}x \right)^2 \leqslant \int_a^x |f'|^2 \, \mathrm{d}x \int_a^x 1^2 \, \mathrm{d}x = (x-a) \int_a^b |f'|^2 \, \mathrm{d}x. \tag{8}$$

Hence

$$\int_a^b f^2 \, \mathrm{d}x \leqslant \int_a^b |f'|^2 \, \mathrm{d}x \int_a^b x - a \, \mathrm{d}x = \frac{(b-a)^2}{2} \int_a^b |f'|^2 \, \mathrm{d}x. \tag{9}$$

9.3.10

Calculate the derivatives of F:
(1) $F(x) = \int_0^{\cos x^2} \sqrt{1+t^2} \, \mathrm{d}t$; (2) $F(x) = \int_{\sin x}^x \, \mathrm{d}t / \sqrt{1+\sin^2 t}$; (3) $F(x) = \int_a^b \sin(x+t) \, \mathrm{d}t$.
Solution: (1)

$$F'(x) = -2x \sin x^2 \sqrt{1+\cos^2 x^2}. \tag{10}$$

(2)

$$F'(x) = \sqrt{1+\sin^2 x} - \cos x / \sqrt{1+\sin^2 \sin x}. \tag{11}$$

(3)

$$F'(x) = \frac{d}{dx} \int_{x+a}^{x+b} \sin t \, \mathrm{d}t = \sin(x+b) - \sin(x+a). \tag{12}$$

9.3.12

Suppose $f \in C(0, \infty)$. If

$$\int_0^x f(t)\,\mathrm{d}t = \frac{1}{2}xf(x), \forall x > 0, \tag{13}$$

prove that there is a constant c such that $f(x) = cx$.
Proof: Let $F(x) = \int_0^x f\,\mathrm{d}t$, then $F(x) = \frac{1}{2}xf(x)$. $f(x) = 2F(x)/x$ is differentiable, so $f(x) = F'(x) = \frac{1}{2}f(x) + \frac{1}{2}xf'(x)$ and $f(x) = xf'(x)$. Hence $\frac{d}{dx}\frac{f(x)}{x} = 0$ so $f(x) = cx$ for some constant c.

9.3.14

Suppose f is convex on $[a, b]$, prove that $f \in \mathcal{R}([a, b])$ and

$$(b-a)f\left(\frac{a+b}{2}\right) \leqslant \int_a^b f\,\mathrm{d}x \leqslant (b-a)\frac{f(a)+f(b)}{2}. \tag{14}$$

Proof: f is convex implies f is continuous (differentiable) except for finitely many points, and f is bounded by $f(a), f(b)$, so $f \in \mathcal{R}([a, b])$. Note that if $x + y = a + b$ then $f(x) + f(y) \geqslant 2f((a + b)/2)$ and $f(x) + f(y) \leqslant f(a) + f(b)$, hence

$$2(b-a)f\left(\frac{a+b}{2}\right) \leqslant 2\int_a^b f\,\mathrm{d}x \leqslant (b-a)(f(a)+f(b)). \tag{15}$$

9.4

9.4.1

(5)

$$\int_0^1 x\sqrt{1-x}\,\mathrm{d}x = \int_0^1 (1-x)\sqrt{x}\,\mathrm{d}x = \frac{2}{3} - \frac{2}{5} = \frac{4}{15}. \tag{16}$$

9.4.2

(10)

$$\int_0^1 \frac{\sqrt{1+x^2}-1}{x^2}\,\mathrm{d}x = \int_0^{\pi/2} \frac{1/\cos^2\theta - 1}{\sin^2\theta}\,\mathrm{d}\theta = \int_0^{\pi/2} \frac{1}{\cos\theta} - \frac{1}{1+\cos\theta}\,\mathrm{d}\theta = \log\left(\sqrt{2}+1\right) - \sqrt{2} + 1. \tag{17}$$

9.4.3

Suppose $0 < r < 1$, prove that

$$\int_0^{2\pi} \frac{1-r^2}{1-2r\cos x + r^2}\,\mathrm{d}x = 2\pi. \tag{18}$$

Proof: The Poisson kernel

$$K_r(x) = \frac{1-r^2}{1-2r\cos x + r^2} = 1 + 2\sum_{n\geqslant 1} r^n \cos nx. \tag{19}$$

Since $K_r(x)$ converges uniformly and $\int_0^{2\pi} \cos nx\,\mathrm{d}x = 0$,

$$\int_0^{2\pi} K_r(x)\,\mathrm{d}x = 2\pi. \tag{20}$$

9.4.4

Suppose $n \in \mathbb{N}$, prove that

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^n)\sqrt[n]{1+x^n}} = 2^{-1/n}. \tag{21}$$

Proof: Let $u = 1 + x^{-n}$, then

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^n)^{(n+1)/n}} = \int_0^1 x^{-(n+1)}(1+x^{-n})^{-(n+1)/n}\,\mathrm{d}x = \frac{1}{n}\int_2^\infty u^{-(n+1)}\,\mathrm{d}u = 2^{-1/n}. \tag{22}$$

9.4.5

(3)

$$\int_{-\pi}^{\pi} \sin mx \cos nx\,\mathrm{d}x = \frac{1}{2}\int_{-\pi}^{\pi} \sin(m+n)x + \sin(m-n)x\,\mathrm{d}x = 0. \tag{23}$$

9.4.6

(9)

$$\int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x}\,\mathrm{d}x = \int_0^{\pi/2} \sum_{k=1}^{n} \frac{\sin(2k-1)\pi}{\sin x}\,\mathrm{d}x = \int_0^{\pi/2} \sum_{k=1}^{n}\sum_{j=1}^{k} \sin jx\,\mathrm{d}x = \frac{n\pi}{2}. \tag{24}$$

9.4.7

Suppose $f \in C([0,1])$, prove that

$$\int_0^\pi x f(\sin x)\,\mathrm{d}x = \frac{\pi}{2}\int_0^\pi f(\sin x)\,\mathrm{d}x, \tag{25}$$

and calculate $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,\mathrm{d}x$.
Proof: Note that $(\pi - x)f(\sin(\pi - x)) + xf(\sin x) = \pi f(\sin x)$, hence the identity holds.

$$\int_0^\pi x \cdot \frac{\sin x}{2-\sin^2 x}\,\mathrm{d}x = \frac{\pi}{2}\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,\mathrm{d}x = \frac{\pi^2}{4}. \tag{26}$$

9.4.8

Suppose f is monotonically decreasing on $(0,\infty)$, denote

$$F(x) = \frac{1}{x}\int_0^x f\,\mathrm{d}t, \forall x > 0 \tag{27}$$

Prove that F is also decreasing on $(0,\infty)$.
Proof: For any $x < x + h$,

$$x(x+h)(F(x) - F(x+h)) = h\int_0^x f\,\mathrm{d}x - x\int_x^{x+h} f\,\mathrm{d}x > hxf(x) - hxf(x) = 0. \tag{28}$$

Hence $F(x) > F(x+h)$.

9.4.9

Suppose f is of period T and $f \in \mathcal{R}(0,T)$, prove that

$$\lim_{\lambda\to\infty}\frac{1}{\lambda}\int_0^\lambda f\,\mathrm{d}x=\frac{1}{T}\int_0^T f\,\mathrm{d}x. \tag{29}$$

Proof: Suppose $M=\int_0^T|f|\,\mathrm{d}x, I=\int_0^T f\,\mathrm{d}x$, and let $n_\lambda=\lfloor\lambda/T\rfloor$, then

$$\left|\frac{1}{\lambda}\int_0^\lambda f\,\mathrm{d}x-\frac{I}{T}\right|\leqslant\left|\frac{In_\lambda}{\lambda}-\frac{I}{T}\right|+\frac{M}{\lambda}\leqslant\frac{I+M}{\lambda}\to 0. \tag{30}$$

Hence

$$\lim_{\lambda\to\infty}\frac{1}{\lambda}\int_0^\lambda f\,\mathrm{d}x=\frac{1}{T}\int_0^T f\,\mathrm{d}x. \tag{31}$$

9.4.10

Suppose $f\in C^2([a,b])$, prove that

$$\int_a^b f\,\mathrm{d}x=\frac{1}{2}(f(a)+f(b))(b-a)+\frac{1}{2}\int_a^b(x-a)(x-b)f''(x)\,\mathrm{d}x. \tag{32}$$

Proof: By integration by parts,

$$\int_a^b(x-a)(x-b)f''\,\mathrm{d}x=-\int_a^b f'\,\mathrm{d}(x-a)(x-b)=-\int_a^b(2x-a-b)\,\mathrm{d}f \tag{33}$$

$$=2\int_a^b f\,\mathrm{d}x-(f(a)+f(b))(a-b). \tag{34}$$

9.4.11

Suppose $f\in C([0,1])$, prove that

$$\lim_{n\to\infty}n\int_0^1 x^n f\,\mathrm{d}x=f(1). \tag{35}$$

Proof: Note that

$$\left|f(1)-(n+1)\int_0^1 x^n f(x)\,\mathrm{d}x\right|\leqslant(n+1)\int_0^{1-\delta}x^n|f(x)-f(1)|\,\mathrm{d}x+(n+1)\int_{1-\delta}^1 x^n|f(x)-f(1)|\,\mathrm{d}x \tag{36}$$

$$\leqslant 2M(1-\delta)^{n+1}+(1-(1-\delta)^{n+1})\varepsilon(\delta). \tag{37}$$

Let $n\to\infty$ we know the limsup of LHS is less then $\varepsilon(\delta)=\sup_{x\in[1-\delta,1]}|f(x)-f(1)|$ for all $\delta>0$. Since $\varepsilon(\delta)\to 0$ as $\delta\to 0$,

$$\lim_{n\to\infty}n\int_0^1 x^n f\,\mathrm{d}x=f(1). \tag{38}$$

9.4.12

Suppose $f\in\mathcal{R}([0,1])$, prove that

$$\lim_{n\to\infty}\int_0^1\sqrt[n]{x}f(x)\,\mathrm{d}x=\int_0^1 f\,\mathrm{d}x. \tag{39}$$

Proof: Suppose $|f|$ is bounded by M, then

$$\left|\int_0^1 \sqrt[n]{x} f(x)\,\mathrm{d}x - \int_0^1 f\,\mathrm{d}x\right| \leqslant M\int_0^1 1-\sqrt[n]{x}\,\mathrm{d}x = M\left(1-\frac{n}{n+1}\right) \to 0. \tag{40}$$

Hence

$$\lim_{n\to\infty}\int_0^1 \sqrt[n]{x} f(x)\,\mathrm{d}x = \int_0^1 f\,\mathrm{d}x. \tag{41}$$

9.4.13

Prove that

$$\int_0^1 \frac{\mathrm{d}x}{1+x^n} = 1-\frac{\log 2}{n}+o\left(\frac{1}{n}\right). \tag{42}$$

Proof: Note that

$$1-\int_0^1 \frac{\mathrm{d}x}{1+x^n} = \int_0^1 \frac{x^n}{1+x^n}\,\mathrm{d}x = \frac{1}{n}\int_0^1 x\,\mathrm{d}\log(1+x^n) = \frac{\log 2}{n} - \frac{1}{n}\int_0^1 \log(1+x^n)\,\mathrm{d}x \tag{43}$$

And

$$\int_0^1 \log(1+x^n)\,\mathrm{d}x \leqslant \int_0^1 x^n\,\mathrm{d}x = \frac{1}{n+1}. \tag{44}$$

Hence

$$\int_0^1 \frac{\mathrm{d}x}{1+x^n} = 1-\frac{\log 2}{n}+o\left(\frac{1}{n}\right). \tag{45}$$

9.4.14

Suppose $f \in C([-1,1])$, prove that

$$\lim_{h\to 0^+}\int_{-1}^1 \frac{hf(x)}{h^2+x^2}\,\mathrm{d}x = \pi f(0). \tag{46}$$

Proof: Let $K_h(x) = h/(h^2+x^2)$ then

$$\int_{-1}^1 K_h(x)\,\mathrm{d}x = 2\arctan\frac{1}{h} \to \pi, \tag{47}$$

and

$$\int_{|x|>\delta} K_h(x)\,\mathrm{d}x < \pi - 2\arctan\frac{\delta}{h}. \tag{48}$$

Hence K_h is sort of a good kernel. Note that

$$\left|\int_{-1}^1 \frac{hf(x)}{h^2+x^2}\,\mathrm{d}x - \pi f(0)\right| \leqslant \left(\pi - 2\arctan\frac{1}{h}\right)|f(0)| + \int_{-\delta}^{\delta} K_h(x)|f(x)-f(0)|\,\mathrm{d}x \tag{49}$$

$$+\int_{|x|>\delta} K_h(x)|f(x)-f(0)|\,\mathrm{d}x \tag{50}$$

$$\leqslant \left(\pi - 2\arctan\frac{1}{h}\right)|f(0)| + \pi\varepsilon(\delta) + 2M\left(\pi - 2\arctan\frac{\delta}{h}\right). \tag{51}$$

Where $\varepsilon(\delta) = \sup_{x\in[-\delta,\delta]} |f(x) - f(0)|$ and $M = \sup_{x\in[-1,1]} |f(x)|$. Let $\delta = \sqrt{h}$ then the right hand side tends to zero as $h \to 0^+$. Therefore

$$\lim_{h\to 0^+} \int_{-1}^{1} \frac{hf(x)}{h^2 + x^2}\,\mathrm{d}x = \pi f(0). \tag{52}$$

9.5

9.5.2

Suppose $f \in \mathcal{R}([a,b])$, and let $F(x) = \int_a^x f(t)\,\mathrm{d}t$, prove that

$$\int_a^b F(x)\,\mathrm{d}x = \int_a^b (b-x)f(x)\,\mathrm{d}x. \tag{53}$$

Proof: We show that $\int_a^x F(t)\,\mathrm{d}t = \int_a^x (x-t)f(t)\,\mathrm{d}t$ for all $x \in [a,b]$: the case $x = a$ is trivial. Taking derivatives on both sides we obtain $F(x) = \int_a^x f(t)\,\mathrm{d}t$ which is clearly true. Hence the identity holds.

9.5.4

Prove that
(1) $\lim_{n\to\infty} \int_0^{\pi/2} \sin^n x\,\mathrm{d}x = 0$:

$$\left|\int_0^{\pi/2} \sin^n x\,\mathrm{d}x\right| \leqslant \frac{\pi}{2}\sin^n\left(\frac{\pi}{2} - \delta\right) + \delta \to \delta \tag{54}$$

for all $\delta > 0$, hence $\int_0^{\pi/2} \sin^n x\,\mathrm{d}x \to 0$.
(2) $\lim_{n\to\infty} \int_n^{n+p} \frac{\cos x}{x}\,\mathrm{d}x = 0$:

$$\left|\int_n^{n+p} \frac{\cos x}{x}\,\mathrm{d}x\right| \leqslant \int_n^{n+p} \frac{1}{x}\,\mathrm{d}x = \log\left(1 + \frac{p}{n}\right) \to 0. \tag{55}$$

9.5.6

Suppose $f(x) = \int_x^{x+1} \sin t^2\,\mathrm{d}t$, prove that $|f(x)| < \frac{1}{x}$ when $x > 0$.
Proof: Note that

$$f(x) = \int_x^{x+1} \frac{-1}{2t}\,\mathrm{d}\cos t^2 = \frac{\cos x^2}{2x} - \frac{\cos(x+1)^2}{2(x+1)} - \int_x^{x+1} \frac{\cos t^2}{2t^2}\,\mathrm{d}t. \tag{56}$$

Hence

$$|f(x)| < \frac{1}{2x} + \frac{1}{2(x+1)} + \frac{1}{2x} - \frac{1}{2(x+1)} = \frac{1}{x}. \tag{57}$$

9.5.8

Suppose $f \in C([a,b])$, prove that

$$\lim_{n\to\infty} \left(\int_a^b |f|^n\,\mathrm{d}x\right)^{1/n} = \max_{x\in[a,b]} |f(x)|. \tag{58}$$

Proof: Clearly $\left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} \leqslant \max_{x\in[a,b]}|f(x)|$. Suppose $f(x_0) = M = \sup_{x\in[a,b]} |f(x)|$ and $x_0 \neq b$, for any $\varepsilon > 0$ take $\delta > 0$ such that $f(y) > M - \varepsilon \forall y \in [x_0, x_0 + \delta]$, then

$$\left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} \geqslant \delta^{1/n}(M - \varepsilon) \to M - \varepsilon \tag{59}$$

for all $\varepsilon > 0$, hence $\lim_{n\to\infty} \left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} = M$.

9.5.9

Suppose $f, g \in \mathcal{R}([a,b])$ and g has period T, prove that

$$\lim_{\lambda\to\infty} \int_a^b f(x)g(\lambda x)\,\mathrm{d}x = \frac{1}{T}\int_a^b f\,\mathrm{d}x \int_0^T g\,\mathrm{d}x. \tag{60}$$

Proof: The case $f = \chi_{[c,d]}$ is proved in Exercise9.4.9, so the identity holds for step functions.
For any $\varepsilon > 0$, there is a step function ϕ such that $\int_a^b |f - \phi|\,\mathrm{d}x < \varepsilon$. Let $I = \int_0^T g\,\mathrm{d}x, M = \sup|g|$, then

$$\left|\int_a^b f(x)g(\lambda x)\,\mathrm{d}x - \frac{I}{T}\int_a^b f\,\mathrm{d}x\right| \leqslant \left(M + \frac{I}{T}\right)\int_a^b |f - \phi|\,\mathrm{d}x + \delta(\lambda) < \varepsilon\left(M + \frac{I}{T}\right) + \delta(\lambda). \tag{61}$$

where $\delta(\lambda) = \left|\int_a^b \phi(x)g(\lambda x)\,\mathrm{d}x - \frac{I}{T}\int_a^b \phi\,\mathrm{d}x\right|$. Let $\lambda \to \infty$ then the lim sup is less then $\varepsilon(M + I/T)$ for any $\varepsilon > 0$, hence it is zero and the identity holds for arbitrary f.

9.6

9.6.6

Suppose $f \in C^1([a,b])$ and $|f'| \leqslant M$, prove that

$$\left|\int_a^b f\,\mathrm{d}x - \frac{f(a) + f(b)}{2}(b - a)\right| \leqslant \frac{1}{4}M(b - a)^2. \tag{62}$$

Proof: Note that

$$\int_a^b f\,\mathrm{d}x - \frac{f(a) + f(b)}{2}(b - a) = -\frac{1}{2}\int_a^b (2x - a - b)f'\,\mathrm{d}x \tag{63}$$

and

$$\int_a^b |2x - a - b|\,\mathrm{d}x = \frac{(b - a)^2}{2}. \tag{64}$$

9.6.7

Suppose $f \in C^2([a,b])$, prove that there exists $\xi \in (a,b)$ such that

$$\int_a^b f\,\mathrm{d}x = (b - a)f\left(\frac{a + b}{2}\right) + \frac{1}{24}(b - a)^3 f''(\xi). \tag{65}$$

Proof: Let $c = \frac{a+b}{2}$ and $h = \frac{b-a}{2}$, consider

$$F(t) = \int_{c-t}^{c+t} f \,\mathrm{d}t - 2tf(c) + Kt^3 \tag{66}$$

where K is a constant such that $F(h) = 0$. Clearly F is an odd function, and $F'(0) = 0$ so $0 = F(h) = h^3 F^{(3)}(\eta)$ for some $\eta \in (0, h)$. Hence $\xi = \eta + c$ satisfy the requirements.

9.7

9.7.1

Calculate arc length:
(1) $y = x^{3/2}, 0 \leqslant x \leqslant 4$.

$$\text{length} = \int_0^4 \sqrt{x^2 + x^3} \,\mathrm{d}x = \int_1^5 (t-1)\sqrt{t} \,\mathrm{d}t = \frac{2}{5}(5^{5/2} - 1) - \frac{2}{3}(5^{3/2} - 1). \tag{67}$$

9.7.2

9.7.3

9.7.4

9.7.5

9.7.10